Fiscal Year:	FY 2010	**Task Last Updated:**	FY 10/28/2010
PI Name:	Hlatky, Lynn Ph.D.		
Project Title:	Computational Modeling of Chromosome Aberrations Produced by HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
PI Email:	Lynn.Hlatky@tufts.edu , LynnHlatky@aol.com	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	(617) 789-2997
Organization Name:	GeneSys Research Institute/Tufts University School of Medicine		
PI Address 1:	736 Cambridge St # CBR-112		
PI Address 2:			
PI Web Page:	http://www.cancer-systems-biology.org		
City:	Boston	**State:**	MA
Zip Code:	02135-2907	**Congressional District:**	8
Comments:	PI affiliation name change to Steward Research and Specialty Projects Corporation (2012). PI formerly at Caritas St. Elizabeth's Medical Center, Boston, MA (2005-2011); Dana-Farber Cancer Institute, Boston, MA (prior to 2005). Subsequent name change from Steward Research and Specialty Projects Corporation to GeneSys Research Institute in 2013.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	09/30/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: grant # changed in 2006 when PI moved--new number is NNJ06HA27G; also, end date is now 9/30/2010, per S. Krenek/JSC (3/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hahnfeldt, Philip (Caritas St. Elizabeth's Medical Center, Tufts Univ School of Medicine) Sachs, Rainer (University of California)		
Grant/Contract No.:	NNJ06HA27G ; NNJ04HF42G		
Performance Goal No.:			
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Task Description:	Chromosome aberrations produced by high LET radiations are directly and indirectly relevant to space exploration and risk estimation in a number of different ways. They are important as probes for DNA repair and recombination mechanisms, as biomarkers of dose and radiation quality, as factors in genomic instability and carcinogenesis, and as surrogates for cancer in risk estimation. Quantitative modeling is needed to exploit and augment the rapidly accumulating empirical data on aberration types and frequencies. This project will apply mathematical and computer algorithms to data on aberrations produced by iron ions (56Fe). The analysis will be compared to analyses of damage produced by photons, electrons, or alpha particles, with the goal of assessing signature responses to 56Fe, an important radiation encountered in space. To perform such studies, the team has tied damage endpoints to risk through our customized software and algorithms. These endpoints include: 1. aberration complexity, quantified by cycle structures; 2. apparently and truly incomplete aberrations, related to segment size distributions; 3. aberration spectra as related to chromosome geometry, to DNA double strand break (DSB) clustering depending on radiation track structure, and especially to DSB misrejoining depending on proximity; 4. mFISH color junctions for each specific chromosome pair, related to chromosome-chromosome spatial correlations in an interphase nucleus. Substantial preliminary studies have recently been carried out by our team for each aim. As remarked, we will employ the team's specialized and comprehensive software, including the highly-developed chromosome aberration simulator (CAS) software, for analyzing the aberration spectra encountered. The data to be analyzed will be for human cells in vitro, scored by mFISH or other similar techniques. A number of these data sets are being made available by many of our colleagues worldwide. This will greatly expand the range of suitable data available for analysis. We have also recently devised methods for systematically comparing mFISH data with data obtained by older methods such as FISH or solid staining and with newer methods such as color banding. For Specific Aim 1, we will replace the current approach of using "obligate" (deterministic) cycles with a more accurate computer-based approach involving cycle probabilities. For Aim 2, new methods are being developed that allow systematic comparison of apparently incomplete aberrations with corresponding complete aberrations and corresponding truly incomplete ones, even in the case of very complex aberrations. Under Aim 2 we will also consider size distributions of fragments that have one telomere and one unrejoined free end, for comparison to recent data using pan-telomeric probes; and analyze mixtures of different radiations. Aim 3 will emphasize more realistic computer algorithms for break misrejoining, bringing this aspect of aberration modeling more nearly in line with the very sophisticated models now being used by various groups for assessing the influence of track structure and chromatin geometry on DSB locations. For Aim 4 we plan to apply a method we previously used for low LET data to data for high LET, including iron ions. The method involves testing frequencies of color junction pairs for deviations from randomness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved biodosimetry Carcinogenesis mathematical models
	Introduction: The specific aims of the grant involved mathematical and computer modeling of data on aberrations produced by iron ions compared to aberrations produced by photons, electrons, alpha particles, or silicon ions. Such aberrations are a surrogate endpoint for quantifying the concerns about HZE carcinogenesis during extended space missions, such as that to Mars. The gap addressed was Cancer07:How can systems biology approaches be used to integrate research on the molecular, cellular, and tissue mechanisms of radiation damage to improve the prediction of the risk of cancer and to evaluate the effectiveness of CMs? How can epidemiology data and scaling factors support this approach? The specific aims of the grant were: 1. Characterizing aberration complexity, quantified by cycle structures. Using this complexity to help indicate the DNA damage/repair/misrepair pathways involved. 2. Relating size distributions of unrejoined and misrejoined DNA segments to DNA double strand break (DSB) clustering within the genome caused by individual HZE radiations or by mixtures of radiations. Systematically comparing apparently incomplete, truly incomplete, and complete aberrations. 3. Developing mechanistic computer simulation of aberrations as related to chromosome geometry, to radiation track structure, and especially to DSB misrejoining depending on spatial proximity of chromosome loci. Simulating overall aberration spectra via CAS (chromosome aberration simulator) to help utilize fully all the information contained in a data set for DNA damage analysis or for specifying biomarkers of radiation quality. 4. Statistically analyzing mFISH color junction frequencies -- for human cells there are 231 color pairs involving autosomes and 45 additional pairs involving the X and/or Y chromosome. Using the pairwise frequencies to indicate chromosome-chromosome spatial associations related to the excess production of interchanges, motivated by the association of specific interchanges with cancer. Also, using the color junction frequencies to indicate the radiosensitivity of each chromosome. Chromosome aberrations produced by high LET radiations are directly and indirectly relevant to space exploration and risk estimation in a number of different ways. They are important as probes for DNA repair and recombination mechanisms, as biomarkers of dose and radiation quality, as factors in genomic instability and carcinogenesis, and as surrogates for cancer in risk estimation. Quantitative modeling is needed to exploit and augment the rapidly accumulating empirical data on aberration types and frequencies. We applied mathematical and computer algorithms to data on aberrations produced by iron ions (56Fe). and compared to analyses of damage produced by photons, electrons, or alpha particles. Methods and Results Aim 1:

Task Progress:

For the part of aim number 1 dealing with analyzing DNA damage/repair/misrepair pathways we used mFISH results on radiogenic homologue-homologue dicentrics, including results on the HZE radiation emphasized by NASA, to show that there is a dependence on homology even for mammalian cells in the G0/G1 phase of the cell cycle. There is a highly significant excess of one color (i.e. homologue-homologue) dicentrics compared with the number of such dicentrics expected from two-color (i.e. heterologous) dicentrics and computer models or analytic estimates. This homology dependence could be due to geometric associations between homologues in the interphase nucleus or to homology-dependent misrepair processes. The dependence of the observed excess on LET is not very strong. The results are reported in reference (1), which is attached in the zip file. We carried out the other part of aim 1, on aberration complexity. We compared complexity for iron ion MFISH data with alpha particle data (kindly supplied by M. Cornforth) using the results on incompleteness from aim 2 (below). We found that automated characterization of complexity can deal with cases too complex for eyeball analysis. The results extended those reported in references (2) and (6).

Aim 2:

We found an elegant and unexpected solution for the part of Aim 2 which deals with systematically comparing apparently incomplete, truly incomplete, and complete mFISH aberrations. In brief, the optimal procedure is to identify incompleteness type by looking for mismatches between centromeres and apparent telomeres involved in a particular aberration since in a complete aberration there are two telomeres of each color for each centromere of that color. Once incompleteness type has been characterized, color-by-color, adding “virtual fragments” of appropriate colors gives a fully methodical way to see what the possible completions are. An applet developed during preliminary studies, is already available on the internet at the URL http://radiobiology.math.berkeley.edu/ . It has been updated with new algorithms that give a more realistic description of the probabilities for different complete aberrations that can correspond to a given apparently incomplete mFISH aberration. Previously, apparent incompleteness has been a major confounding factor in analyzing HZE data on aberrations. These results and their application to iron ion data are also reported in reference (6).

We found new results for that part of aim 2 which concerns DSB (DNA double-strand break) clustering at high LET and its influence on misrejoining frequencies. We have collaborated with A. Ponomarev and F. Cucinotta of NASA in work on how to subtract out background values of DNA fragment size distributions in PFGE experiments on radiation breakage to analyze DSB yields and clustering. The group has found a fully general subtraction method, based on stochastic point-process theory, which works for non-random (high LET) double strand break induction as well as random (low LET) double strand break induction and for random or non-random background breakage. The method is also applicable to subtracting out low LET effects on break distributions in the case of mixed HZE radiations and protons on DSB clustering. Moreover it can explain the observed dose dependence of break distribution sizes in our data and that of others mathematically. In brief: as dose increases, extra breaks cut larger segments into smaller ones, which tends to decrease the number of large segments and increase the number of small segments, so that the former number depends grows less than linearly with dose and the latter more than linearly. The results were reported in 2 peer reviewed papers (4) and (8).

Application of Aim 2: We began the mathematical analysis of aberrations which appear during cancer progression, especially in aneuploidy and in the cell-cell fusion as recently observed in our lab. Such aberrations often involve the loss, gain, or replacement (e.g. via loss of heterozygosity) of whole chromosomes or of chromosome segments so large that they are visible or visibly missing optically at metaphase. Mathematically what is required is an extension of the aberration multigraph formalism that relaxes the requirement of conservation of genetic material. We have developed a tentative formalism using a single "reservoir" from which extra chromosome segments are drawn and to which missing segments are relegated, with appropriate initial or final edges in the aberration multigraph imposed.

Aim 3. Simulations and dose dependence:

We carried out Aim 3, computer simulations of aberration formation, specifically the development of Chromosome Aberration Simulator (CAS) software. We developed a geometric approach to high-LET radiation simulation, based on random consecutive location of chromosomes and usimg our randomly located cluster formalism (references 4 and 8).. It was implemented by Java software involving about 30 classes and interfaces. This code is documented in detail at the URL http://math.berkeley.edu/ . It gives detailed dose-response curves for all classes of chromosome aberrations. In addition we are now working on an approach based on random walk geometry for all 46 chromosomes in an interphase nucleus, using track structure simulations, implemented by a simulated annealing approach to locating chromosomes in an interphase nucleus in which overlapping of different chromosome territories is allowed but is subjected to an energy penalty. The results are in better agreement with recent (non-radiobiological) data on chromosome territories.

Application of Aim 3: We utilized dose-dependence for radiation damage formation, linear quadratic at low to moderate doses, as suggested by kinetic studies on aberration formation and corroborated by CAS, to analyze incidence of second cancers after tumor radiotherapy, taking into account cell population dynamics for the heterogeneous cell population during and after fractionated protocols and stochastic effects implied by the fact that in early stages the number of pre-malignant cells can be very small. The results were reported in a series of papers (5, 7, 9, 10).

Aim 4:

We carried out specific aim number 4: statistically analyzing mFISH color junction frequencies for radiation-produced chromosome aberrations. We used the pairwise frequencies to indicate chromosome-chromosome spatial associations related to the excess production of interchanges and also used the color junction frequencies to indicate the radiosensitivity of each chromosome. Data and results on high LET radiation (i.e. densely ionizing radiation such as alpha particles, or HZE particles) were added to the analyses for low LET (i.e. for low linear energy transfer, corresponding to sparsely ionizing radiation such as hard x-rays or gamma rays). The high LET data is particularly pertinent to the kind of radiation encountered in an extended space voyage. We found that proximity effects on interchanges due to geometric associations between chromosomes in an interphase nucleus are more pronounced at high LET.

A previously suggested chromosome cluster in lymphocytes consisting of 5 gene-rich chromosomes was confirmed and a cluster containing the nucleolus chromosomes was confirmed. Chromosome associations and preferential radiogenic interchanges due to proximity effects are important in radiation carcinogenesis, especially radiation leukemogenesis. The

results on specific aim number 4 were reported in reference (3).

Discussion

We conclude the following:

* There is a straightforward way, analogous to the microdosimetric method of predicting multi-track action from single-track action but slightly more sophisticated, to interpret data on DNA fragment sizes after low or high LET radiation. The results suggest that for the fluences encountered in GCR, carcinogenic damage is probably additive when dealing with a mixture of radiations.

* The problem of interpreting incomplete aberrations, which is troublesome especially at high LET, has a systematic solution, which should make the databases on such aberrations more useful.

* The usual, primarily statistical, methods of estimating GCR carcinogenesis (e.g. from atomic bomb survivor data combined with scaling methods) should be supplemented by more mechanistic modeling which can better take into account the possibility that radiation acts on promotion, transformation and progression, not just initiation.

Peer-Reviewed Publications Crediting the Grant.

1. Plan Y., Hlatky L., Hahnfeldt P., Sachs R., Loucas B., and Cornforth M. Full-color painting reveals an excess of radiation-induced dicentrics involving homologous chromosomes. Int J Radiat Biol 81:613-20, 2005.

2. Levy D., Hlatky L. and Sachs R.K. Software for cycle analysis, http://radiobiology.berkeley.edu/ . NSF, 2006.

3. Vives S., Loucas B., Vazquez M., Brenner D.J., Sachs R.K., Hlatky L., Cornforth M., Arsuaga J. SCHIP: Statistics for chromosome interphase positioning. Bioinformatics 21:3181-2, 2005.

4. Ponomarev A.L., Belli M., Hahnfeldt P.J., Hlatky L., Sachs R.K., and Cucinotta F.A. A robust procedure for removing background damage in assays of radiation-induced DNA fragment distributions. Radiat Res 166(6):908-916, 2006.

5. Shuryak I., Sachs R.K., Hlatky L., Little M.P., Hahnfeldt P., and Brenner D.J. Radiation-induced leukemia at doses relevant to radiation therapy: modeling mechanisms and estimating risks. J Natl Cancer Inst 98(24):1794-1806, 2006.

6. Levy D., Reeder C., Loucas B., Hlatky L., Chen A., Cornforth M., Sachs, R. Interpreting chromosome aberration spectra. J Comp Biol 14:144-155, 2007.

7. Sachs R.K., Shuryak I., Brenner D., Fakir H., Hlatky L., Hahnfeldt P. Second cancers after fractionated radiotherapy: stochastic population dynamics effects. J Theor Biol 249:518-531, 2007.

8. Ponomarev, A.L. Hlatky L., Sachs R.K., Hahnfeldt P., Cucinotta F.A. Subtraction of background damage in PFGE experiments on DNA fragment-size distributions. Radiat Environ Biophys 46(2):155-160, 2007.

9. Shuryak I., Hahnfeldt P., Hlatky L., Sachs R.K. and Brenner D.J. A new view of radiation-induced cancer: integrating short- and long-term processes. Part II: second cancer risk estimation. Radiat Environ Biophys 48:275-286, 2009.

10. Shuryak I., Hahnfeldt P., Hlatky L., Sachs R.K. and Brenner D.J. A new view of radiation-induced cancer: integrating short- and long-term processes. Part I: approach. Radiat Environ Biophys 48:263-274, 2009.

Students:

Dan Levy: was supported and obtained PhD at UCB based on extending the work in the two bibliography references.

Allen Chen: was supported and obtained Masters Degree at UCB on topics related to the work.

Y Plan and Chris Reeder contributed to the papers in the bibliography and they subsequently obtained Bachelor's degrees.

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Shuryak I, Hahnfeldt P, Hlatky L, Sachs RK, Brenner DJ. "A new view of radiation-induced cancer: integrating short- and long-term processes. Part I: approach." Radiat Environ Biophys. 2009 Aug;48(3):263-74. PubMed PMID: 19536557 , Aug-2009
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Articles in Peer-reviewed Journals	Beheshti A, Benzekry S, McDonald JT, Ma L, Peluso M, Hahnfeldt P, Hlatky L. "Host age is a systemic regulator of gene expression impacting cancer progression." Cancer Res. 2015 Mar 15;75(6):1134-43. https://doi.org/10.1158/0008-5472.CAN-14-1053 ; PubMed PMID: 25732382; PubMed Central PMCID: PMC4397972 , Mar-2015
Articles in Peer-reviewed Journals	Girdhani S, Sachs R, Hlatky L. "Biological effects of proton radiation: an update." Radiat Prot Dosimetry. 2015 Sep;166(1-4):334-8. Epub 2015 Apr 20. https://doi.org/10.1093/rpd/ncv178 ; PubMed PMID: 25897135 , Sep-2015